

```
clear all; clc; close all;  
s=tf('s')
```

s =

s

Continuous-time transfer function.

```
m=1;  
b=3;  
k=2;  
A=[0 1; -k/m -b/m]
```

A = 2×2

0	1
-2	-3

```
B=[0; 1/m];  
C=[1 0];  
D=0;  
Id=[1 0; 0 1];  
lambdas=eig(A)
```

lambdas = 2×1

-1
-2

```
[N,D]=ss2tf(A,B,C,D)
```

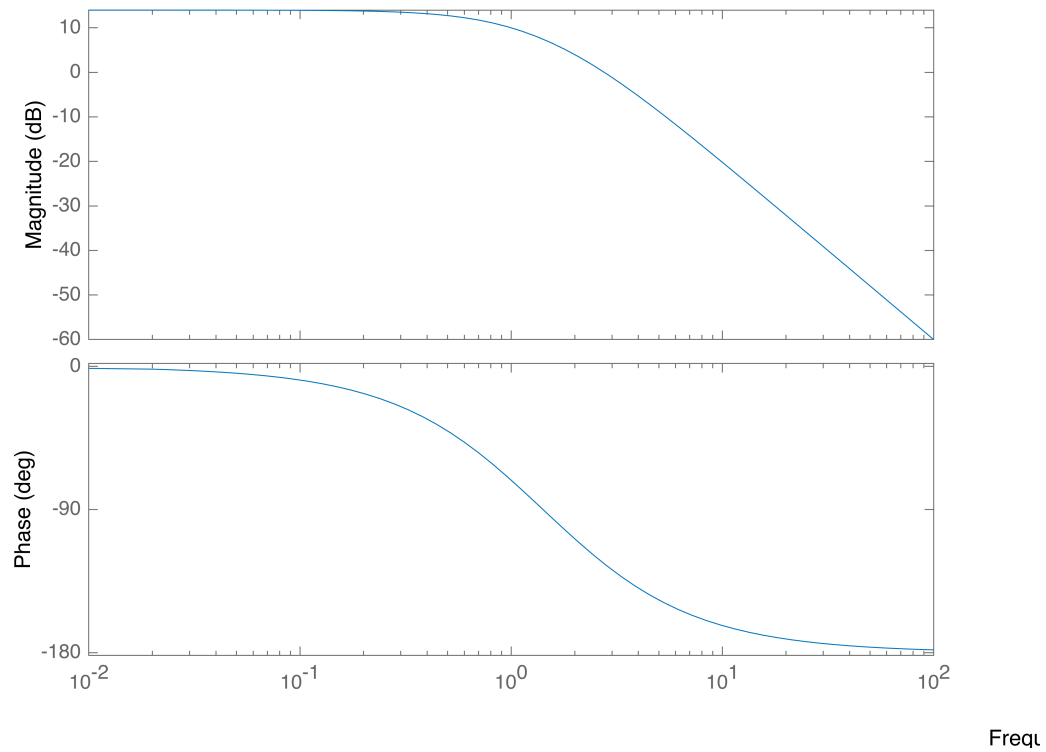
N = 1×3

0	0	1
---	---	---

D = 1×3

1	3	2
---	---	---

```
G=1/(s^2+3*s+2);  
bode(10*G)
```



```
W= feedback(10*G,1)
```

W =

$$\frac{10}{s^2 + 3s + 12}$$

Continuous-time transfer function.

```
roots([1 3 12])
```

```
ans = 2×1 complex
-1.5000 + 3.1225i
-1.5000 - 3.1225i
```

```
step(W)
```

